

Sizhe Li

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EDUCATION

Johns Hopkins University

Incoming Ph.D. in Computer Science

Research Interests: Theoretical Computer Science, Online Algorithms, Learning-Augmented Algorithms.

Maryland, USA

Sep. 2026 –

University of California, Berkeley

Visiting Student

GPA Rank: 1st/600+

California, USA

Aug. 2024 – Dec. 2024

The Chinese University of Hong Kong, Shenzhen

Bachelor of Science in Statistics

Major GPA Rank: 1st/42 Overall GPA Rank: 2nd/42

Shenzhen, China

Sep. 2022 – Jun. 2026

PUBLICATIONS

- **Sizhe Li**, Nicolas Christianson, Tongxin Li. *Prediction-Specific Design of Learning-Augmented Algorithms*. **ACM SIGMETRICS 2026**.
- **Sizhe Li**, Nicolas Christianson, Tongxin Li. *Near-Optimal Consistency-Robustness Bounds for Non-Clairvoyant Scheduling*. **Under Review at SODA 2027**.

RESEARCH EXPERIENCE

Prediction-Specific Design of Learning-Augmented Algorithms

Feb 2024 – May 2025

Supervised by **Prof. Tongxin Li** and **Prof. Nicolas Christianson**

- **New Framework & Definitions:** Introduced a *prediction-specific framework* for learning-augmented algorithms enabling fine-grained, per-prediction performance guarantees. Proposed a novel definition of *strong optimality* capturing Pareto optimality in both consistency and robustness tradeoffs. Proved the suboptimality of state-of-the-art methods (NeurIPS 2018, 2021) in this new framework.
- **Generic Approach:** Developed a *bi-level optimization* framework for designing strongly-optimal algorithms, applicable to diverse online problems.
- **Explicit Algorithm Design:** Designed provably strongly optimal algorithms for *deterministic ski rental*, *randomized ski rental* and *one-max search*.

Optimal Tradeoff in Learning-Augmented Non-Clairvoyant Scheduling

June 2025 – Present

Supervised by **Prof. Tongxin Li** and **Prof. Nicolas Christianson**

- This work advances a central open problem in learning-augmented online scheduling: determining the optimal consistency–robustness trade-off for general n jobs. Prior bounds (Wei & Zhang, NeurIPS’20) were tight only for $n = 2$ and two extreme points. We prove that the classical lower bound is not tight for $n \geq 5$, establish refined lower and upper bounds for finite n along with the matching algorithm, and derive analytical lower and upper bounds in the asymptotic regime as $n \rightarrow \infty$, providing a near-optimal characterization of the frontier. In particular, our results close the $\Theta(\lambda^{-1/2})$ gap in the worst-case n regime.

TEACHING

Undergraduate Teaching Fellow, CUHK-Shenzhen

- MAT2041: Linear Algebra and Applications Fall 2023, Spring 2025
- CSC1001: Introduction to Computer Science: Programming Methodology Fall 2025
- STA4001: Stochastic Processes Spring 2026

HONORS AND AWARDS

- Selected for *Top Scholar Recruitment Fellowship*, University of Washington, Department of Electrical and Computer Engineering, 2026 (Declined)
- Undergraduate Research Award, 2025–2026
- ACM SIGMETRICS Travel Grant, 2026
- Outstanding Graduate, School of Data Science, CUHK-SZ, 2026
- Academic Performance Scholarship (*Top 1%*), 2023–2024
- Dean's List, 2023–2025

SKILLS

- **Algorithms and Theory:** Approximation algorithms, competitive analysis, complexity theory
- **Programming:** Python (NumPy, Pandas, SciPy, Matplotlib), Java, R, MATLAB
- **Tools:** Git, Linux, Jupyter, VS Code, Overleaf
- **Typesetting:** $\text{\LaTeX}$